

Alex John Bush

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EMPLOYMENT HISTORY

MAS Medical Assurance Society

From **September 2024** to **now**

Leading the architecture **design and delivery of data-driven technology transformation**, including the introduction of a **near real-time operational data store**, future-focused approach to **system integration and cross-system data mastery**, and heavily involved in the wider **technology transformation programme** including supporting both a **CRM and Insurance Policy Administration System (PAS) migration**. Supported the Data team in general, including **acting Head of Data**, Lead Data Engineer support, mentoring of junior team members, **platform security and infrastructure hardening**, and **driving best practices in data engineering and DevOps**.

Solutions & Data Architect

Professional Services, Qrious Limited (NZ)

From **November 2022** to **September 2024**

Led delivery of many modern data platforms and data products across a range of technologies and government and private sector clients. Involved with significant amount of **pre-sales activities**, from **solution architecture design**, to **project scoping, costing and proposal writing**. **Managed a team of data consultants**, from junior to senior consultants. **Drove automation and repeatable infrastructure-as-code initiatives** to bring innovation to project delivery, ensure platform implementation consistency and capture reusable assets.

Principal Consultant

Led the **architecture design and platform delivery** of **Microsoft Fabric**-based modern data platform for multiple large organisations. Led the architecture design and platform delivery of **Azure Databricks** for analyst workloads in a **large NZ government organisation**. Delivered multiple modern data platform implementations across **Fivetran, Azure, Snowflake, Azure Databricks, Coalesce and DBT** using **Infrastructure-as-Code**.

Digital Lighthouse, KPMG New Zealand (NZ)

From **August 2019** to **November 2022**

Led and delivered **Data and Analytics** projects for **public and private sector clients**. **Drove internal innovation to increase delivery capability and technical maturity**. **Fostered a culture of learning and collaboration through mentoring**.

Associate Director

From **April 2022** to **November 2022**

Chapter Lead for Engineering. Led **business development** and **engagement delivery**, and identified and delivered **strategic investments in technology and mentoring** to ensure **delivery capability and capacity**. Managed relationships with senior client stakeholders to ensure **effective delivery of complex engagements**.

Senior Manager

From **April 2021** to **March 2022**

Solution design and delivery of end-to-end **analytics platforms** for public-sector clients. Delivered **zero-trust platform architectures** of PaaS and IaaS infrastructure. Developed **large-scale data processing pipelines in Apache Spark** to support client data modelling.

Manager

From **August 2019** to **March 2021**

Led client project delivery, from Azure Cloud architecture and infrastructure automation, to back-end application development and data reporting. Increased the **technical maturity of the team**, running **workshops and designing best practices for delivery**.

Cox Automotive (UK)

From **November 2016** to **August 2019**

Data Engineering team lead, supporting Business Intelligence (BI) and Data Science (DS) teams with data and platforms to **enable effective business analytics**. Enabled principle of **self-service analytics** and ETL development for BI and DS teams (summarised in a **talk given at the Spark Summit 2019** (youtu.be/1WUlua-xjJA)).

Lead Data Engineer

From **April 2018** to **August 2019**

Delivered and supported **data-driven platforms, pipelines, and products** across BI and DS teams. Performed **zero-downtime platform migration** from **Cloudera (IaaS) to Azure Databricks (PaaS)**. Was a core contributor to **Waimak**, an open-source Apache Spark library (github.com/CoxAutomotiveDataSolutions/waimak).

Hadoop DevOps Engineer

From **November 2016** to **March 2018**

Automated deployment of Cloudera platform, and **migrated platform onto a secure platform** (Kerberos, SSL, RBAC). Responsible for all **automation of project and environment builds and deployment**.

Hortonworks (UK)

From **December 2015** to **November 2016**

Professional services consultant providing Hadoop **implementation, administration and developer** services to banking-sector clients.

Senior Consultant

British Gas (UK)

From **September 2013** to **November 2015**

Developer and Platform Engineer delivering **ETL pipeline development**, and **infrastructure and platform automation** to support large on-premises **Hadoop clusters**.

EDUCATION

MPhys, Masters in Computational Physics with Honors
University of Edinburgh
Masters Project Radiative Transfer on GPUs using NVIDIA CUDA
Senior Honors Project Lattice QCD on GPUs using NVIDIA CUDA

CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert
Microsoft
Valid from February 2022 to February 2025

Microsoft Certified: Azure Security Engineer Associate
Microsoft
Valid from September 2020 to September 2023

Microsoft Certified: DevOps Engineer Expert
Microsoft
Valid from January 2022 to January 2025

Microsoft Certified: Azure Developer Associate
Microsoft
Valid from February 2024 to February 2025

Microsoft Certified: Azure Data Engineer Associate
Microsoft
Valid from April 2024 to April 2026

Microsoft Certified: Azure Administrator Associate
Microsoft
Valid from January 2022 to January 2025

Microsoft Certified: Fabric Analytics Engineer Associate
Microsoft
Valid from May 2024 to May 2025

SnowPro Core Certification
Snowflake
Valid from November 2023 to November 2025

SnowPro Advanced: Architect Certification
Snowflake
Valid from August 2024 to August 2026

SKILLS

Data Engineering
Apache Spark, Databricks, Snowflake, Microsoft Fabric, dbt, and SQL

Software Engineering
Scala, Python, C#, TypeScript, and Functional Programming

Cloud Engineering
Azure, and AWS

DevOps / Automation

Azure DevOps, ARM / Bicep, GitHub, and Terraform

Infrastructure / Platform
Linux / Unix, Bash, Networking, and Dynamics 365

Soft Skills
Solution Architecture and Platform Design, Agile Project Delivery, Requirements Gathering, Technical Writing, Workshop Facilitation, and Data Modelling